EncCluster: Scalable Functional Encryption in Federated Learning through Weight Clustering and Probabilistic Filters

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Pervasive and Mobile Computing, 2025

01 Context & approach

The challenge

Encrypting every model weight makes secure federated aggregation expensive. *EncCluster* encrypts a small set of representative centroids and compresses the position mapping, reducing client encryption work and transmitted data.

Our contribution

Combine weight clustering with decentralized multi-client functional encryption.
Encode weight-to-centroid assignments using Binary Fuse filters.
Evaluate accuracy, upload volume, and encryption time across four architectures.

1 Train and cluster

Train locally, then group similar weights into a compact set of centroids and record their assignments.

2 Encrypt the compact update

Encrypt centroid values. Encode position-cluster pairs with a Binary Fuse filter and send them with partial functional keys.

3 Reconstruct and aggregate

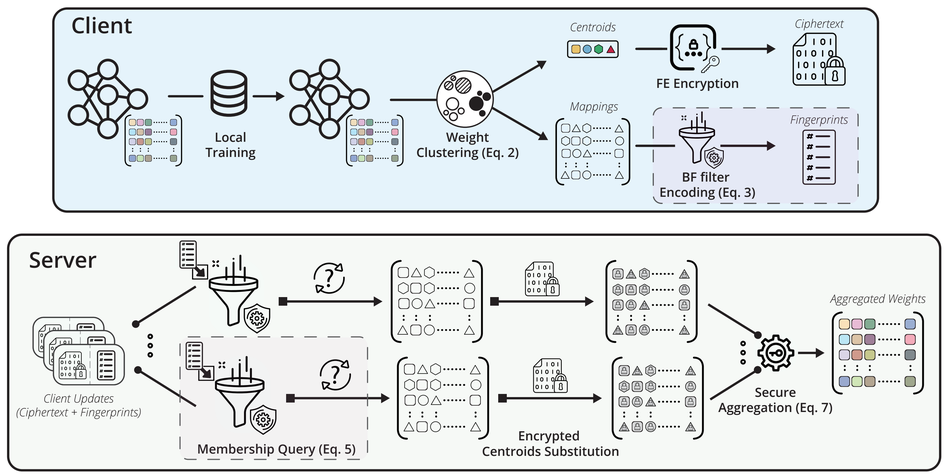
Recover the mapping, expand encrypted centroids into model positions, and decrypt the authorized weighted aggregate.

Scope

Assumes an honest-but-curious setup authority/server and secure channels; poisoning and denial-of-service are outside scope. Filter errors do not provide differential privacy. Simulation timings are hardware-specific; repeated-run uncertainty is unreported.

02 Method

Encrypt compact centroids, then reconstruct encrypted weights for secure aggregation.



Core formulation

$$\mathcal{L}_{\text{cluster}} = \sum_{i=1}^{d} (\theta_i - z_{a_i})^2$$

θ_i: weight; z: centroids; a_i: assigned cluster. Approximate many weights with a few shared values before encryption.

$$\theta_j^{r+1} = \frac{1}{|D|} \text{Dec}(\{\widehat{\theta}_{n,j}^{r}\}_n, \text{dk}_f)$$

Encrypted weights are preweighted by client sample count. Divide the decrypted sum by total samples |D|; dk_f authorizes aggregation.

Paper

Code

03 Results

0.284×
Upload volume / FedAvg
CIFAR-10 · full participation

2.04 s
Client encryption time
A10 / 96-core cluster simulation

Lower encrypted communication and computation

Method	IID Δ (pp)	Non-IID Δ (pp)	Upload / FedAvg	Encrypt (s)
FedAvg + clustering	-0.12	-0.71	0.034	—
DeTrust-FL	-0.09	-0.10	3.743	329.53
SEFL	-0.98	-1.74	2.581	7.64
EncCluster (Ours)	**-0.32**	**-0.79**	**0.284**	**2.04**

CIFAR-10 · ResNet-20 · 30 clients · full participation · 100 rounds. Dir(10)/Dir(0.1), 128 centroids, 256-bit keys. Deltas vs. FedAvg; timings from an A10 / 96-core cluster simulation.

Architecture coverage

Architecture	C10 Δ (pp)	C100 Δ (pp)	Encrypt (s)
ConvNeXt-Tiny	-0.36	-1.29	2.02
ResNet-20	-0.32	-1.19	2.03
MobileNet	-0.24	-0.97	2.08
ConvMixer-256/8	-0.19	-1.02	2.12

IID CIFAR-10/100 · 10 clients · full participation · 100 rounds. Accuracy changes use each architecture's FedAvg baseline; same cluster hardware.

Takeaway

EncCluster reduces uploads to **0.284× FedAvg** with a **0.32-point IID accuracy decrease** on CIFAR-10. Client encryption takes **2.04 s** in the A10 / 96-core cluster simulation, versus 7.64 s for SEFL.